Experiment ‘p_6_2_i_has_adjacent_duplicates_seq’ Results

June 4, 2026

Experiment outcome: SUCCESS
Bad responses: 0
Responses containing `assume` : 0
Responses containing `expect` : 0
Resolution attempts: 1
Hard fails (resolution): 0
Soft fails (resolution): 0
Verification attempts: 1

Problem Specification

Problem name: p_6_2_i_has_adjacent_duplicates_seq
Natural language statement: Write a method to return true if the given array contains two adjacent duplicate elements.
Method signature: p_6_2_i_has_adjacent_duplicates_seq(inputs: seq<int>) returns (has_duplicates: bool)

Ensures

- `has_duplicates == (exists i :: 0 <= i < |inputs| - 1 && inputs[i] == inputs[i + 1])`

Requires

- `|inputs| >= 2`

GenAI interactions

Below you will find all interactions between the 'user' (program) and the 'assistant' (OpenAI).

Program → GenAI

```
You are given the following task to perform in Dafny:

Write a method to return true if the given array contains two adjacent duplicate elements
    ↪ .

The signature should be:

p_6_2_i_has_adjacent_duplicates_seq(inputs: seq<int>) returns (has_duplicates: bool)

The method should respect the following contract:

requires |inputs| >= 2, ensures has_duplicates == (exists i :: 0 <= i < |inputs| - 1 &&
    ↪ inputs[i] == inputs[i + 1])

Produce and show only the Dafny body of this method, including the curly braces that
    ↪ surround it. Do not show the signature nor contract. You must not use 'assume' nor
    ↪  'expect' anywhere in your code.
```

GenAI → Program

System fingerprint: null
ID: chatcmpl-Dn53kgK6vj0yiMVd0BwBDNlgUy6Wd
Model: gpt-5.5-2026-04-23
Created at: 1780589892
Finish reason: stop
Usage: 330 tokens in, and 667 tokens out

```
{
  var i := 0;
  while i < |inputs| - 1
    invariant 0 <= i <= |inputs| - 1
    invariant forall j :: 0 <= j < i ==> inputs[j] != inputs[j + 1]
  {
    if inputs[i] == inputs[i + 1] {
      has_duplicates := true;
      assert exists j :: 0 <= j < |inputs| - 1 && inputs[j] == inputs[j + 1];
      return;
    }
    i := i + 1;
  }
  has_duplicates := false;
}
```

Final Program

```
method p_6_2_i_has_adjacent_duplicates_seq(inputs: seq<int>) returns (has_duplicates:
    ↪ bool)
        requires |inputs| >= 2
        ensures has_duplicates == (exists i :: 0 <= i < |inputs| - 1 && inputs[i] ==
            ↪ inputs[i + 1])
{
  var i := 0;
  while i < |inputs| - 1
    invariant 0 <= i <= |inputs| - 1
    invariant forall j :: 0 <= j < i ==> inputs[j] != inputs[j + 1]
  {
    if inputs[i] == inputs[i + 1] {
      has_duplicates := true;
      assert exists j :: 0 <= j < |inputs| - 1 && inputs[j] == inputs[j + 1];
      return;
    }
    i := i + 1;
  }
  has_duplicates := false;
}
```

Total Token Usage

Input tokens: 330
Output tokens: 667
Reasoning tokens: 512
Sum of 'total tokens': 997

Experiment Timings

Overall Experiment started at 1780589879175, ended at 1780589933567, lasting 54392ms (54.39 seconds)
Iteration #1 started at 1780589880366, ended at 1780589933567, lasting 53201ms (53.20 seconds)